Why Do Computer Networks Exist?

A computer network is a system that enables multiple computing devices to communicate and share information with one another. While a single computer can perform computations, run applications, and store data, its true potential is realized only when it can interact with other devices. Networking exists because computing becomes significantly more useful and powerful when machines are able to exchange information and collaborate.

A Powerful Computer Can Still Be Isolated

Imagine having an extremely powerful computer capable of running advanced applications, executing Large Language Models (LLMs), playing locally stored media, and performing complex computations without requiring an Internet connection. From a computational perspective, the machine is highly capable. However, despite all its power, it has one major limitation: it operates in isolation.

If a computer is not connected to any network, applications running on it remain accessible only on that local machine.

Thousands or even millions of computers around the world may host valuable services, databases, and applications.

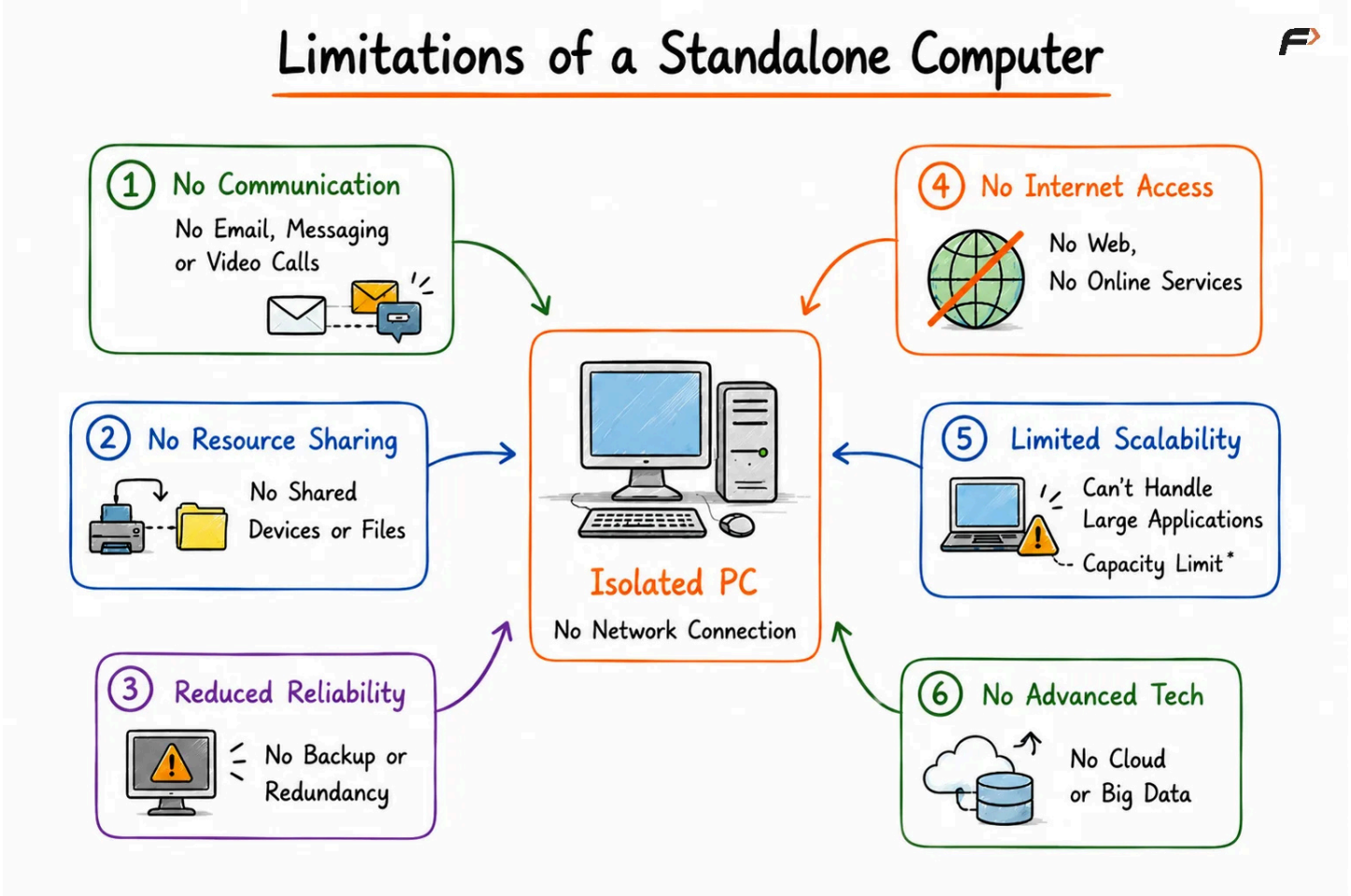
Without communication between these computers, users would not be able to access services hosted on other systems.

Information, files, and resources would remain isolated within individual devices.

Networking enables computers to exchange data and make remote services available to users anywhere in the world.

The modern Internet is essentially a massive network that allows billions of devices to communicate and share services with one another.

This is exactly why computer networks exist. **A powerful computer without a network is similar to a highly intelligent person locked alone in a room**. The capabilities are present, but there is no mechanism to share knowledge, communicate, or collaborate with others.



Why Machines Need to Communicate

Modern computing is built around communication between devices. Most applications that we use daily depend on machines exchanging information across a network. Without networking, many technologies that we take for granted simply would not exist.

Some common examples include:

Sending messages through WhatsApp

Sharing files using Google Drive

Accessing websites and web applications

Sending and receiving emails

Participating in video conferences

Playing online multiplayer games

Using cloud-based applications and services

Consider a few of these examples. When you send a WhatsApp message, the message must travel from your device to another user's device. When you upload a file to Google Drive, that

file is transferred to remote servers. Similarly, whenever you access a website, information is exchanged between your device and the server hosting the website. All of these interactions rely on networking.

The fundamental purpose of networking can therefore be summarized as follows:

Enable communication between devices

Allow information sharing

Provide access to remote services and applications

Support collaboration between users and systems

Enable large-scale distributed computing

Networking Does Not Always Mean Internet

One of the most common misconceptions among beginners is assuming that a network and the Internet are the same thing. In reality, a network is a much broader concept. A network refers to any collection of devices that can communicate with each other, regardless of whether they are connected to the Internet.

The relationship can be summarized as:

Every Internet connection involves networking.

Not every network requires Internet connectivity.

A network can exist entirely within a room, building, or organization.

The Internet is simply the world's largest interconnected network.

Examples of Networks Without Internet

Many networks operate perfectly well without any connection to the public Internet. The only requirement is that the participating devices should be able to communicate with one another.

AirPods and a Smartphone

Consider a pair of AirPods connected to a smartphone through Bluetooth. The phone continuously sends audio data to the AirPods, while the AirPods communicate status information such as battery levels and connection details back to the phone. Since these devices are exchanging information, they form a network even though no Internet connection is involved.

Laptop Connected to Wi-Fi Without Internet

You may have encountered situations where your laptop is connected to a Wi-Fi network but the Wi-Fi itself does not have Internet access. Even in such cases, the laptop remains part of a network because communication with other devices connected to the same network is still possible.

Office Networks

Organizations commonly maintain internal networks consisting of computers, printers, servers, and other devices connected through switches and cables. Such networks allow users to:

Share files

Access shared printers

Communicate with internal servers

Use common organizational resources

Even if the organization's Internet connection fails, these services can continue operating within the local network.

Servers Inside a Data Center

A data center is a facility containing a large number of interconnected servers. These servers continuously exchange information with one another to support applications, cloud services, and websites. Most of this communication happens through private networking infrastructure within the data center itself.

The Internet becomes necessary only when these servers need to communicate with users or systems located outside the data center.

What Is the Internet Then?

If networks can exist independently of the Internet, then what exactly is the Internet? The Internet is a global system that connects multiple networks together. For this reason, it is often described as a **network of networks**.

For example, consider two data centers located in different parts of the world. Each data center has its own internal network of servers. If these networks need to exchange information across continents, a larger communication infrastructure is required. This worldwide interconnection is provided by the Internet.

In simple terms:

A network connects devices.

The Internet connects networks.

Network vs Internet

Network	Internet
A collection of devices that can communicate with each other.	A global collection of interconnected networks.
Can be very small in size.	Operates on a worldwide scale.
May function without Internet access.	Connects multiple independent networks together.
Examples include Bluetooth connections, office LANs, and data center networks.	Examples include websites, cloud services, and online communication systems.

Conclusion

Computer networks exist because computing becomes truly useful when machines can communicate with one another. A standalone computer may be powerful, but its capabilities remain limited if it cannot exchange information, share resources, or provide services to other systems.

Whether it is a Bluetooth connection between a phone and AirPods, an office LAN, a data center network, or the global Internet, the underlying goal remains the same: enabling devices to communicate and share information. The Internet is simply the largest example of networking, connecting millions of individual networks together across the world.

Types of Networks

Computer networks can be classified based on the geographical area they cover and the scale at which they operate. Some networks are designed to connect devices around a single person, while others connect systems across cities, countries, and even continents. Understanding these network types is important because they form the foundation of modern communication systems.

The major types of networks are:

Personal Area Network (PAN)

Local Area Network (LAN)

Wireless Local Area Network (WLAN)

Campus Area Network (CAN)

Metropolitan Area Network (MAN)

Wide Area Network (WAN)

As we move from PAN to WAN, the geographical coverage increases and the network becomes larger and more complex.

1. Personal Area Network (PAN)
2. PAN stands for Personal Area Network. It is the smallest type of network and is designed to connect devices around a single person. The primary purpose of a PAN is to enable communication between personal devices over a very short distance.

A PAN typically operates within a range of a few meters and is commonly used for connecting devices that belong to the same user.

Some common examples of PAN are:

Smartphone connected to wireless earbuds

Smartphone connected to a smartwatch

Mobile hotspot shared with nearby devices

The main characteristics of a PAN are:

Very small coverage area

Limited to a single user

Low power consumption

Does not necessarily require Internet connectivity

Used for personal communication between devices

2. Local Area Network (LAN)

LAN stands for Local Area Network. It connects devices within a small geographical area such as a home, office, one floor of a building, or a college computer laboratory.

Devices located in the same place often need fast communication and access to shared resources. A LAN makes this possible by providing a high-speed network within a limited area.

Consider an office containing laptops, desktop computers, printers, and other devices. If all these devices are connected and can communicate with each other, they form a LAN.

Common examples of LAN include:

Home Wi-Fi network

Office network

College computer lab network

Key features of a LAN are:

Small geographical coverage

High-speed communication

Low latency

Shared Internet access

Privately managed network

3. Wireless Local Area Network (WLAN)

A WLAN, or Wireless Local Area Network, is essentially a LAN that uses Wi-Fi instead of physical cables. The purpose and coverage area remain the same; only the communication medium changes.

For example, imagine an office with ten computers. If those computers are connected using network cables, the setup is considered a LAN. However, if the same computers are connected through Wi-Fi, the network becomes a WLAN.

In simple terms:

LAN uses wired connections.

WLAN uses wireless connections such as Wi-Fi.

4. Campus Area Network (CAN)

CAN stands for Campus Area Network. It connects multiple LANs within a campus, college, university, corporate office park, or any similar organizational area.

A single campus may contain multiple buildings, departments, laboratories, libraries, and hostels. Each building may have its own LAN, and a CAN is used to interconnect all of these LANs into a larger network.

Examples of CAN include:

University campuses

College campuses

Corporate office parks

Large organizational premises

A CAN is larger than a LAN but smaller than a city-wide network.

5. Metropolitan Area Network (MAN)
5. MAN stands for Metropolitan Area Network. It connects multiple LANs across a city or a large urban area. As the name suggests, a MAN operates at a metropolitan scale and is larger than a LAN or CAN.

Suppose an organization has multiple offices located in different parts of a city. Each office may have its own LAN, and a MAN can be used to connect all these LANs together.

A simple example would be:

Office LAN 1 located in one part of the city

Office LAN 2 located in another part of the city

A city-wide network connecting both offices

Telecommunication providers such as Airtel, Jio, and other service providers commonly use MAN infrastructure to provide connectivity across urban areas.

A MAN is:

Larger than a LAN

Larger than a CAN

Smaller than a WAN

6. Wide Area Network (WAN)

WAN stands for Wide Area Network. It covers very large geographical areas such as multiple cities, states, countries, or even continents. WANs are designed to connect networks that are separated by large distances.

For example, a network in New York may need to communicate with another network in London, or a network in London may need to communicate with one in Tokyo. Such long-distance communication is made possible through WAN infrastructure.

WANs typically rely on telecommunication infrastructure such as fiber-optic networks and undersea communication cables to connect networks across the world.

Some common examples of WAN include:

The Internet

AWS regions connected across countries

Google Cloud regions connected across countries

One important concept to remember is that a WAN is usually formed by connecting multiple smaller networks together over long distances.

For example, an AWS data center in Mumbai and another AWS data center in New York may communicate through WAN infrastructure. In this case, WAN acts as the bridge connecting networks separated by thousands of kilometers.

Key characteristics of WAN are:

Covers very large geographical areas

Connects multiple smaller networks

Uses telecom infrastructure

Often relies on fiber-optic and undersea cables

Has higher latency compared to LANs

Network Scale Comparison

One of the easiest ways to remember the different network types is to arrange them according to their scale.

Network Type Coverage

PAN Around a single person

LAN Home, office, or computer lab

WLAN Wireless version of a LAN

CAN Campus or office park

MAN City or large urban area

WAN Countries and continents

As we move from PAN to WAN, the geographical coverage continuously increases. PAN is the smallest network type, while WAN is the largest and is responsible for connecting networks across the world.

Conclusion

Computer networks are classified according to the area they cover. PAN connects devices around a single user, LAN and WLAN provide local connectivity, CAN connects multiple LANs within a campus, MAN connects networks across a city, and WAN connects networks across countries and continents. Together, these network types form the building blocks of modern communication systems and help us understand how connectivity scales from personal devices to global infrastructure.

The Internet

The Internet is a global system of interconnected computer networks that communicate with one another using the TCP/IP protocol suite. The most important phrase in this definition is **interconnected networks**. The Internet is not a single network, nor is it owned or controlled by a single company. Instead, it is formed by connecting together thousands of independently owned networks across the world.

This interconnected structure allows devices, organizations, cloud providers, universities, telecom companies, and Internet Service Providers (ISPs) to communicate seamlessly with one another. Whether you are accessing a website, sending a message, uploading a file, or streaming a video, your data travels through multiple interconnected networks before reaching its destination.

Internet: A Network of Networks

A common way to describe the Internet is as a **network of networks**. Individual networks can exist independently, but when they are interconnected and allowed to communicate with one another, they collectively form the Internet.

For example, consider an AWS region located in Mumbai and another AWS region located in New York. These regions communicate with each other using networking infrastructure such as fiber-optic links and undersea cables. Similarly, when a user sitting in Bengaluru accesses an application hosted inside an AWS region, the communication involves not only AWS infrastructure but also multiple other networking entities such as Internet Service Providers.

This highlights an important fact:

The Internet is not owned by a single company.

Different parts of the infrastructure are owned by different organizations.

Communication occurs across multiple interconnected networks.

Together, these networks form the global Internet.

Who Contributes to the Internet?

The Internet is built and maintained by numerous organizations, each operating its own portion of the network infrastructure. These independent networks cooperate with one another to enable global communication.

Some major contributors include:

Internet Service Providers (ISPs) such as Airtel and Jio.

Cloud Providers such as AWS, Azure, and Google Cloud.

Telecom Companies that provide networking infrastructure.

Universities and Research Institutions operating campus networks.

Enterprises and Organizations maintaining private corporate networks.

Since data often travels through infrastructure owned by multiple organizations, a common communication standard is required. This is where TCP/IP becomes important.

The Role of TCP/IP

If different networks owned by different organizations need to communicate, they must all understand a common language. The Internet uses the **TCP/IP protocol suite** as that common language.

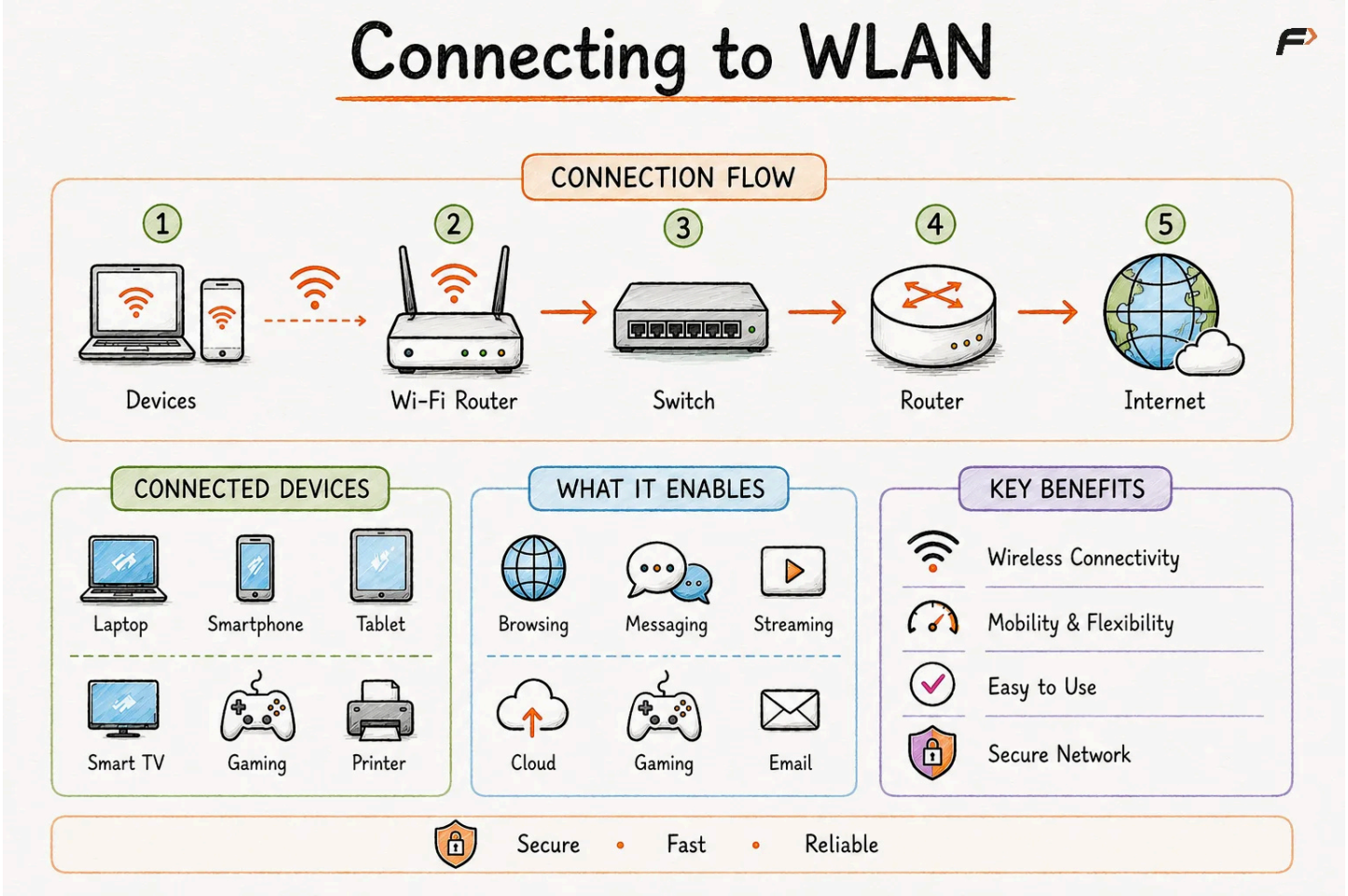
A useful analogy is human communication. People from different backgrounds can communicate effectively if they share a common language. Similarly, cloud providers, ISPs, telecom companies, universities, and enterprise networks can exchange data because they all follow the TCP/IP protocol suite.

TCP/IP defines the rules that allow devices and networks to communicate, exchange data, and understand each other's messages regardless of who owns the underlying infrastructure.

How Does Your Home Connect to the Internet?

Whenever you access a website, send a message, or use an online service, your request follows a sequence of networking components before reaching its destination.

The typical communication flow is:



Step 1: Device to Home Router

Your laptop, smartphone, or other device connects to a home router using either Wi-Fi or a LAN cable. The home router acts as the gateway between your local network and the Internet.

Step 2: Home Router to ISP

The home router forwards data to your Internet Service Provider through a physical communication medium such as a fiber-optic cable.

Inside the fiber-optic cable, data is converted into light pulses. These light pulses travel through the fiber at extremely high speeds and are converted back into digital data at the receiving end.

Step 3: ISP to the Internet

Once the request reaches the ISP router, the ISP determines where the data needs to go. The destination could be a WhatsApp server, a website hosted in AWS, a cloud application, or any other online service.

The ISP then forwards the request through larger Internet infrastructure until it reaches the desired network.

Step 4: Destination Server

Eventually, the request reaches the server hosting the application or service. The server processes the request and sends a response back through a similar path.

Internet Service Providers (ISPs)

An Internet Service Provider (ISP) is an organization that provides users with access to the Internet. ISPs act as the bridge between home networks and the broader Internet infrastructure.

Examples of ISPs include:

Airtel

Jio

BSNL

ACT

Without an ISP, a home network would remain isolated and would not be able to communicate with external networks across the Internet.

Backbone Networks

Backbone Networks form the core infrastructure of the Internet. They are extremely high-capacity networks responsible for carrying enormous volumes of traffic across large geographical distances.

To understand why backbone networks are needed, consider the connection between a home router and an ISP router. The fiber connecting a single home to an ISP carries a relatively limited amount of traffic because it serves only one household. There is no need to install extremely expensive infrastructure for such a connection.

However, once traffic from thousands or millions of users is aggregated together, the amount of data becomes enormous. At that scale, high-capacity backbone infrastructure becomes necessary.

Backbone networks are designed to:

Carry massive amounts of Internet traffic.

Connect major ISPs and data centers.

Connect countries and continents.

Provide high-speed long-distance communication.

Ensure reliable and redundant connectivity.

Highway Analogy for Backbone Networks

A useful way to understand backbone networks is through a transportation analogy.

Your home network is like a local street.

Your ISP network is like a city road.

The Internet backbone is like a multi-lane express highway.

Small roads are sufficient for local traffic, but long-distance travel between cities requires large highways capable of handling heavy traffic efficiently. Similarly, backbone networks act as the Internet's express highways, carrying huge volumes of data between major networks across the world.

Key Takeaways

The Internet is a global system of interconnected networks.

No single company owns the Internet.

ISPs, cloud providers, telecom companies, universities, and enterprises all contribute to the Internet.

TCP/IP acts as the common language used for communication.

Home devices connect to the Internet through routers and ISPs.

Fiber-optic cables transmit data using light pulses.

Backbone networks form the high-capacity core infrastructure of the Internet.

Backbone networks enable communication across countries and continents.

Clients, Servers and Peers

Communication in computer networks happens through interactions between different systems. Some systems request services, some provide services, and in certain architectures a system can do both. Understanding the concepts of clients, servers, and peers is fundamental because most modern applications, websites, APIs, and communication platforms are built around these communication models.

Client

A client is a device or program that requests a service from another system. It initiates communication whenever it needs data, functionality, or access to a resource. In most networking scenarios, communication begins with a client sending a request.

Some common examples of clients include:

A browser requesting the homepage of a website.

A mobile application calling an API to fetch data.

Postman sending requests to a backend service for testing.

A simple way to remember the role of a client is:

The client always asks.

The client initiates communication.

The client waits for a response.

Server

A server is a device or program that provides a service to requesting systems. It receives requests, processes them, and returns the appropriate response. While clients initiate communication, servers are responsible for serving the requested data or functionality.

Examples of servers include:

A web server returning HTML content for a website.

An API server returning a JSON response.

A backend service processing application requests.

The role of a server can be summarized as follows:

The server listens for requests.

The server processes incoming requests.

The server sends responses back to clients.

Understanding Client and Server Through Examples

Consider a browser opening a website. The browser sends an HTTP request to a web server, and the web server responds with HTML content. In this interaction, the browser acts as the client and the web server acts as the server.

Similarly, when a mobile application calls an API, the application acts as the client while the API server acts as the server. The application sends a request and the API server responds with data, often in JSON format.

The fundamental principle remains the same:

Client → Sends Request

Server → Sends Response

A Server Can Also Be a Client

A single machine can function as both a client and a server, depending on the direction and nature of the communication.

Consider the following scenario:

A mobile application sends a request to an API server.

The API server needs data from a database.

The API server sends a request to the database.

The database returns the required data.

In the first interaction, the API server behaves as a server because it is responding to the application's request. However, in the second interaction, the same API server behaves as a client because it is requesting data from the database.

This demonstrates an important principle:

The same system can be a client in one communication flow and a server in another.

Client–Server Architecture

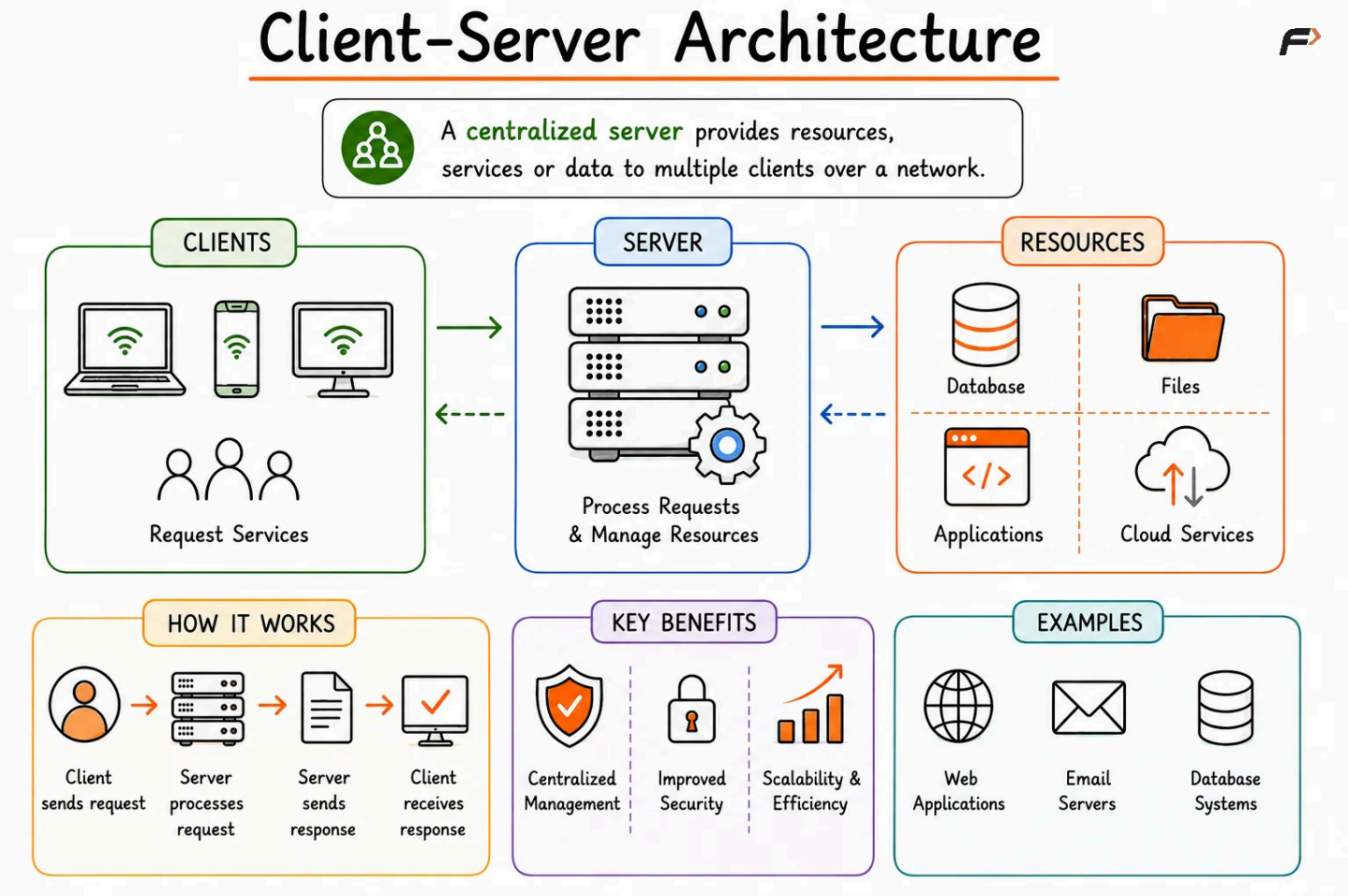
The client-server architecture is one of the most common communication models in computer networking. It is based on a simple principle: clients request services and servers provide them.

In this architecture:

Clients initiate communication.

Servers process requests.

Servers return responses.



Examples of client-server architecture include:

Browser ⇿ Web Server

Mobile App ⇿ API Server

API Server ⇿ Database Server

This architecture exists because it provides easier control and management. Data and business logic can be centralized on servers, making applications easier to maintain and manage.

Peer-to-Peer (P2P) Architecture

Peer-to-Peer (P2P) architecture follows a different approach. Instead of having a dedicated requester and a dedicated provider, every participant in the network can both request and provide services.

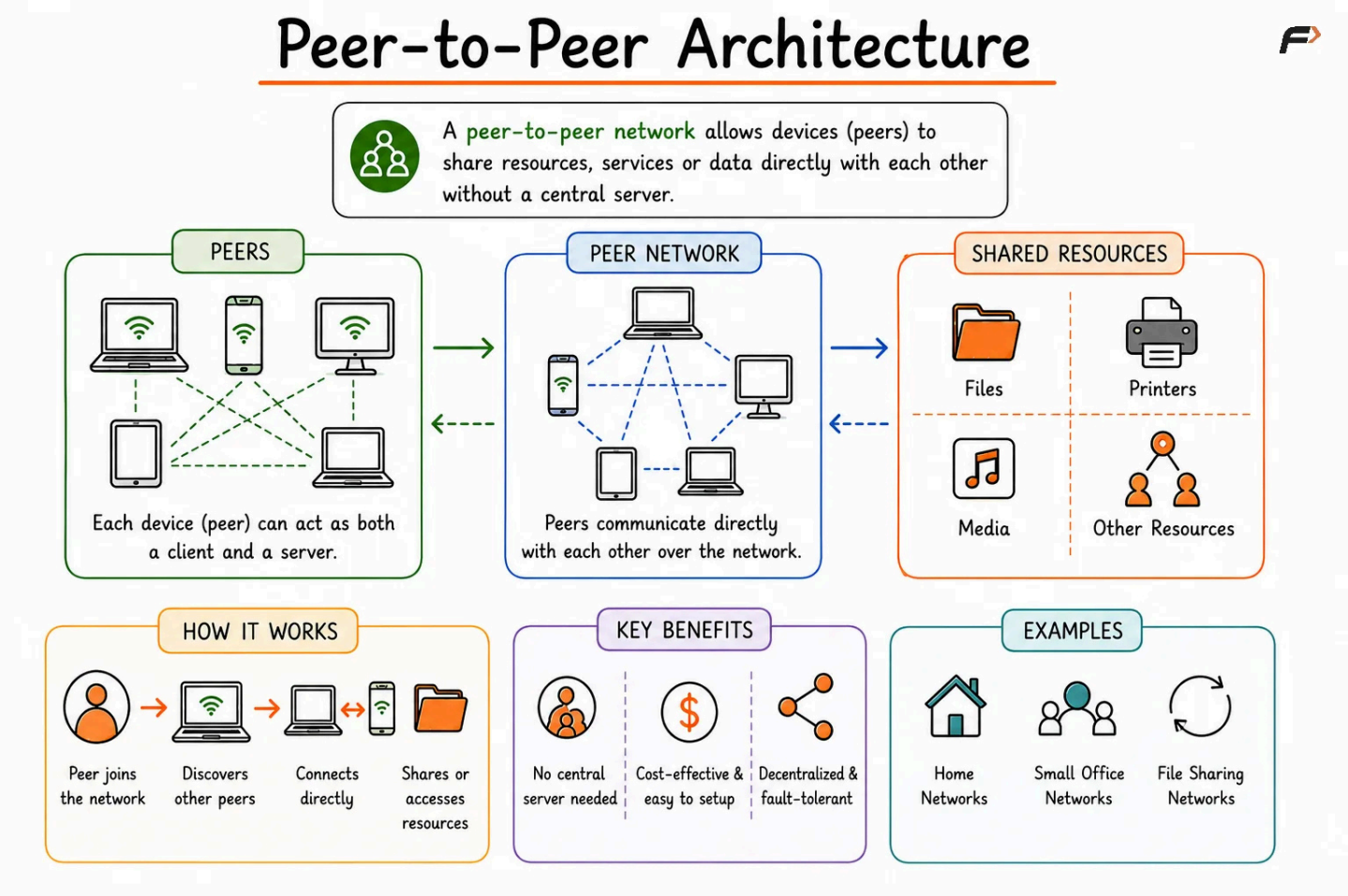
In a P2P network:

Any peer can request data.

Any peer can provide data.

Communication can happen simultaneously.

There is no strict separation between client and server.



Unlike client-server systems where communication is centered around a server, peer-to-peer systems are more distributed because multiple machines communicate directly with one another.

A peer can request data from another peer while simultaneously serving data to a different peer. This ability to both consume and provide services is what distinguishes peer-to-peer systems from traditional client-server architectures.

Client–Server vs Peer-to-Peer

Client-Server	Peer-to-Peer
Client requests, server responds.	Every peer can both request and provide.
Centralized architecture.	Distributed architecture.
Communication depends on servers.	Peers communicate directly.
Roles are clearly separated.	Roles are shared among peers.

A common example is WebRTC-based communication, where participating systems can exchange data directly with one another using a peer-to-peer model.

Hosts, Nodes and Links

To understand networking infrastructure, it is important to become familiar with three fundamental terms: host, node, and link. These concepts help describe how devices participate in a network and how communication occurs between them.

Host

A host is an end device that sends or receives data in a network. Hosts are typically the devices that directly participate in communication as the source or destination of information.

Examples of hosts include:

Laptops

Smartphones

Servers

Desktop computers

Whether a device is requesting data or responding with data, it can be considered a host if it acts as an endpoint in the communication process.

Node

A node is any device that forms part of the network infrastructure. While hosts are end devices, nodes are often the devices that help data travel through the network.

Examples of nodes include:

Routers

Switches

Firewalls

In a typical communication path, nodes sit between the client and server and help forward, process, or manage network traffic.

Link

A link is the communication path that connects devices within a network. Whenever two devices communicate, some medium is required to carry the data between them. That medium is referred to as a link.

Examples of links include:

Wi-Fi connections

Ethernet cables

Fiber-optic cables

For example, a laptop may be connected to a router through Wi-Fi, while the router may be connected to an ISP through a fiber-optic cable. These communication paths are links.

Router

A router is a networking device responsible for connecting different networks and forwarding data between them. Whenever data needs to travel from one network to another, routers determine the path that the data should follow to reach its destination.

For example, when you access a website from your laptop, the request first reaches your home router. The router then forwards the request toward your Internet Service Provider (ISP), which in turn forwards it through additional routers across the Internet until it reaches the destination server.

Common examples of routers include:

Home Wi-Fi routers connecting local devices to the Internet.

Enterprise routers connecting office networks.

ISP routers forwarding traffic between different networks.

Core Internet routers that help move data across countries and continents.

In a typical Internet communication flow, a router acts as the gateway that allows devices inside one network to communicate with devices located in other networks.

Putting It All Together

Consider a laptop communicating with a server:

The laptop acts as a host.

The server acts as a host.

Routers and switches between them act as nodes.

The communication paths connecting them are links.

In simple terms:

Hosts use the network.

Nodes form the network.

Links connect everything together.

Conclusion

Clients, servers, and peers define how systems communicate in a network. Client-server architecture relies on dedicated requesters and responders, while peer-to-peer architecture allows every participant to act as both. To support this communication, networks are built using hosts, nodes, and links. Hosts generate and receive data, nodes help move data through the network, and links provide the communication paths that connect everything together.

How Data is Transmitted in a Network

When data is sent over a network, it is not transmitted as one large continuous block. Instead, it is divided into smaller units called **packets**. These packets travel through the network independently and are reassembled at the destination to reconstruct the original data.

A simple analogy is transporting the contents of an entire house. Instead of moving everything in one giant shipment, items are divided into smaller labeled packages. These packages are easier to transport, manage, track, and recover if something goes wrong. Networking follows a very similar principle.

Why Data Is Broken Into Packets

Imagine that you want to send a 100 MB file across the Internet. The data travels from your device to a home router, then to an ISP, through backbone networks, and eventually reaches the destination server.

If the entire 100 MB were transmitted as one giant block and the transmission failed midway, the entire file would have to be sent again from the beginning. This would waste time, bandwidth, and network resources.

Breaking data into packets solves this problem and provides several advantages:

Easier Network Sharing: Multiple users can transmit packets simultaneously without blocking one another.

Easier Forwarding: Routers can process and forward smaller units of data more efficiently.

Better Error Recovery: If a packet is lost, only the missing packet needs to be retransmitted.

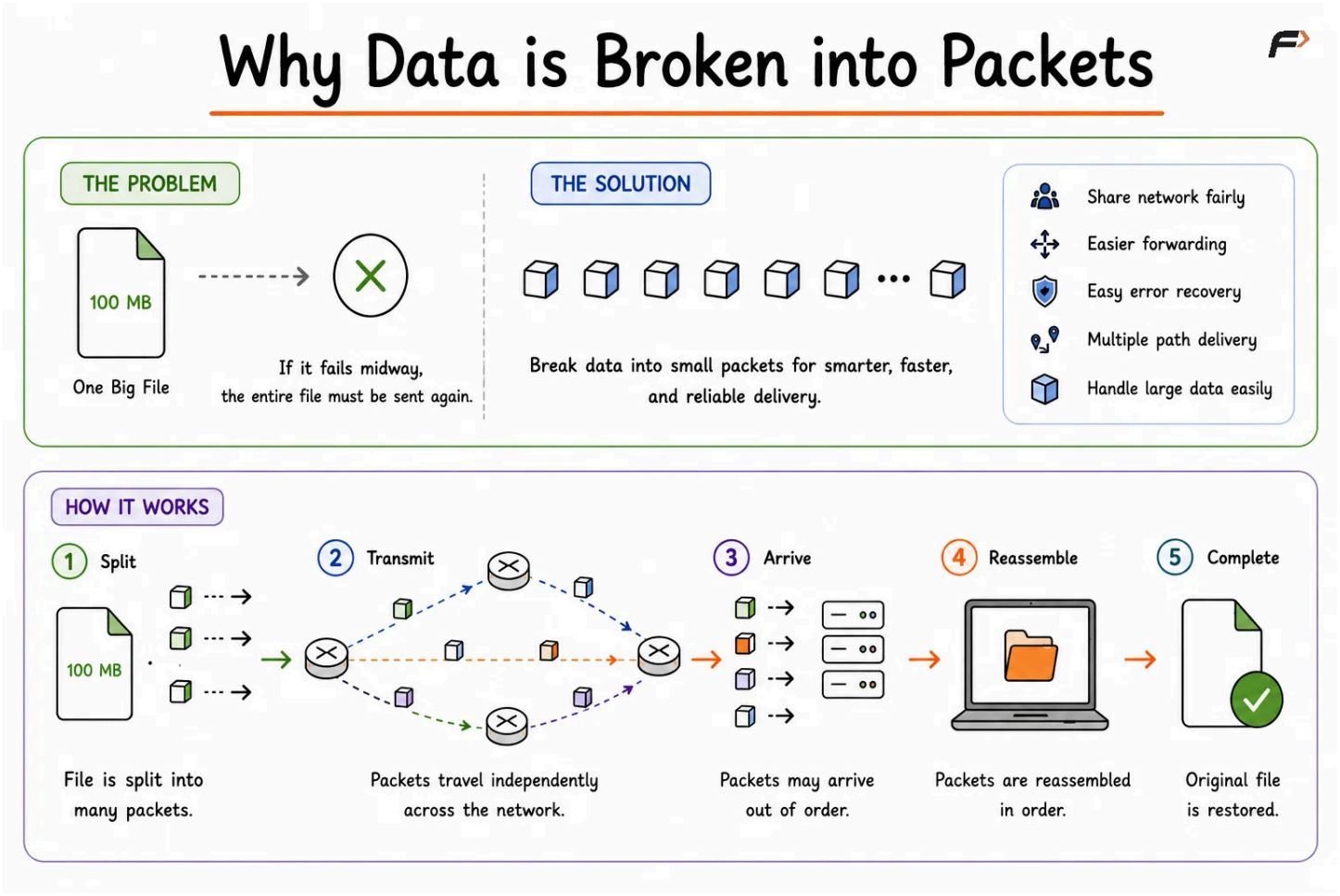
Improved Reliability: Packets can take alternative paths if one route becomes unavailable.

Better Management of Large Data: Large files can be handled in manageable chunks.

Understanding Packets Through an Example

Consider a five-minute audio file. Instead of sending the entire audio file at once, it can be divided into many smaller packets. For example, each packet may contain a few seconds of audio data.

The first packet may contain the first few seconds of audio, the second packet may contain the next few seconds, and so on. Collectively, these packets represent the complete audio file.



This process allows the network to handle large amounts of data efficiently while minimizing the impact of transmission failures.

How Packets Reach the Destination

Once data has been divided into packets, each packet is transmitted independently through the network. Modern networks often provide multiple paths between the source and destination.

As a result:

Packet 1 may travel through one path.

Packet 2 may travel through a different path.

Packet 3 may travel through yet another path.

Even though packets may follow different routes, they eventually arrive at the destination and are reassembled into the original message.

Sequence Numbers and Reassembly

Since packets can arrive out of order, the destination system needs a mechanism to reconstruct the original data correctly. This is achieved using **sequence numbers**.

Each packet carries identifying information that helps determine its position within the original message. When all packets arrive, the receiving system uses these sequence numbers to reassemble the data in the correct order.

Packet Loss and Retransmission

Not every packet successfully reaches its destination. A packet may be lost due to network congestion, transmission errors, hardware failures, or physical issues such as a damaged fiber-optic cable.

Suppose Packet 3 is lost while Packets 1, 2, 4, and 5 successfully reach the destination. Instead of retransmitting the entire file, the network can simply request Packet 3 again. The sender then retransmits only the missing packet.

This approach is significantly more efficient than resending the complete message and is one of the key reasons why packet-based communication is used throughout the Internet.

Key Idea: Packets are small chunks of a larger message that can be transmitted, routed, recovered, and reassembled efficiently.

Important Network Performance Terms

To understand how well a network performs, several important metrics are used. These metrics help measure capacity, speed, responsiveness, reliability, and the effectiveness of data transmission.

Bandwidth

Bandwidth refers to the maximum theoretical capacity of a communication link. It represents the maximum amount of data that can be transmitted through a network connection under ideal conditions.

Bandwidth is usually measured in bits per second (bps).

Examples:

100 Mbps Ethernet connection

1 Gbps fiber connection

10 Gbps backbone link

Think of bandwidth as the maximum capacity that a road or highway can support.

Throughput

Throughput represents the actual rate at which data is successfully transmitted between the sender and receiver. Unlike bandwidth, throughput reflects real-world performance.

Even if a connection has a bandwidth of 100 Mbps, users may only experience 60 Mbps due to factors such as:

Network congestion

Protocol overhead

Retransmissions

Hardware limitations

Environmental conditions

A useful analogy is fuel efficiency. A car manufacturer may advertise a mileage of 20 km/l, but real-world conditions such as traffic and driving style may result in only 15 km/l. Similarly, throughput is often lower than the theoretical bandwidth.

Another analogy is a highway designed to support 100 cars per hour but only carrying 60 cars per hour due to traffic and delays.

Latency (Delay)

Latency is the time required for a packet to travel from the sender to the receiver. It is typically measured in milliseconds (ms).

Low latency results in faster response times and a more responsive user experience, while high latency can lead to noticeable delays.

Latency is especially important for:

Online gaming

Video conferencing

Voice calls

Real-time collaboration tools

In general, lower latency is better.

Packet Loss

Packet loss occurs when one or more packets fail to reach their destination. Lost packets must often be retransmitted, which can reduce performance and degrade user experience.

Packet loss can occur because of:

Network congestion

Faulty hardware

Damaged communication links

Transmission errors

Reliable protocols detect missing packets and request retransmission when necessary.

Goodput

Goodput measures the rate at which useful application-level data is successfully delivered to the destination. Unlike throughput, goodput excludes protocol overhead, retransmissions, and control information.

When data is transmitted through a network, additional information is often attached to each packet. Examples include:

Headers

Sequence numbers

Control information

Protocol metadata

While these additions are necessary for networking, the application itself only cares about the actual user data.

Therefore:

Throughput = Total data transmitted.

Goodput = Useful application data delivered.

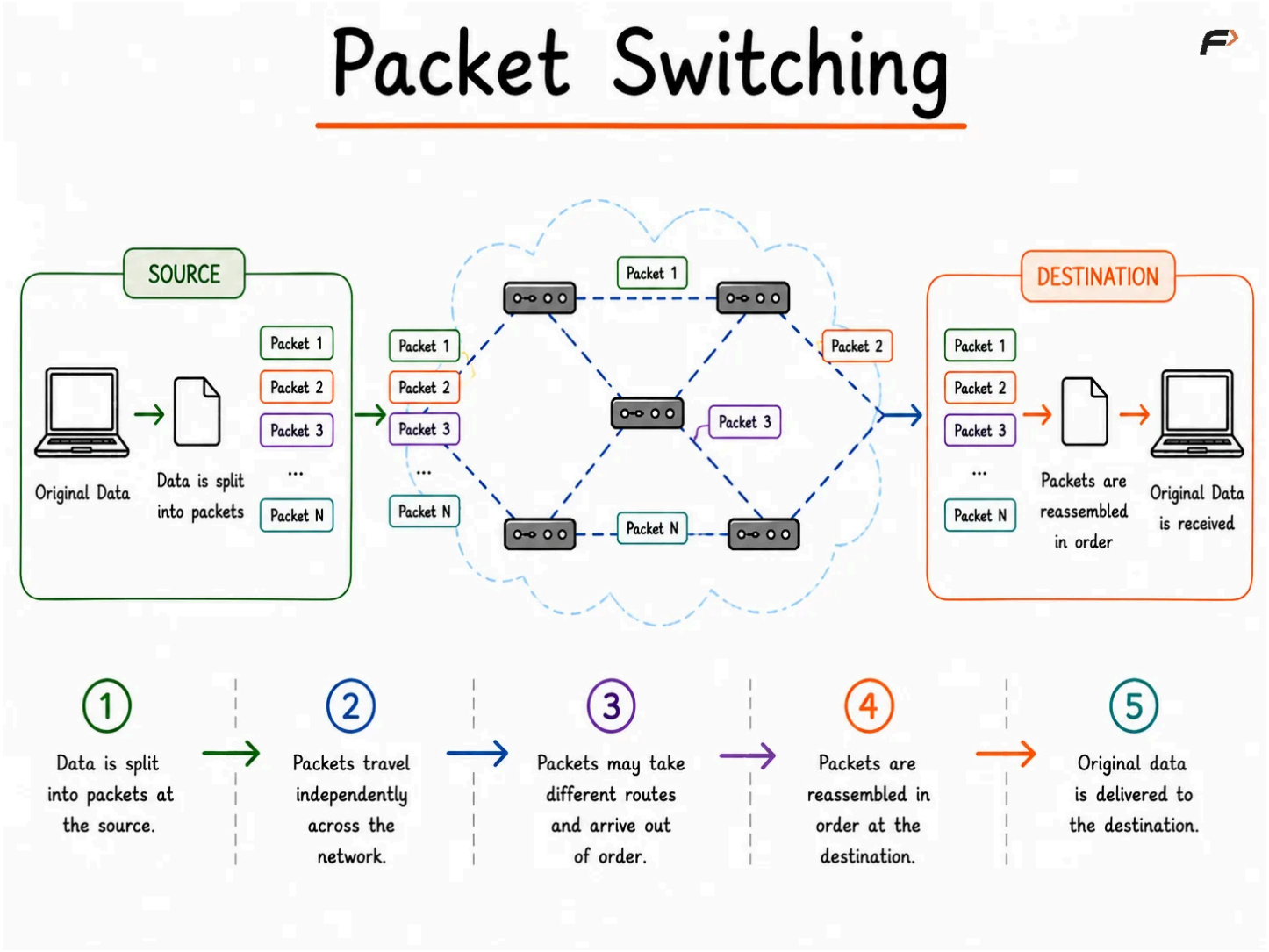
Goodput provides a better representation of the effective data rate experienced by the end user.

Packet Switching vs Circuit Switching

Computer networks use different techniques to transfer data from a source to a destination. The two most fundamental approaches are packet switching and circuit switching.

Packet Switching

Packet switching is the technique used by the Internet. In this approach, data is divided into packets and each packet is transmitted independently through the network.



Key characteristics include:

Data is divided into packets.

No dedicated path is reserved.

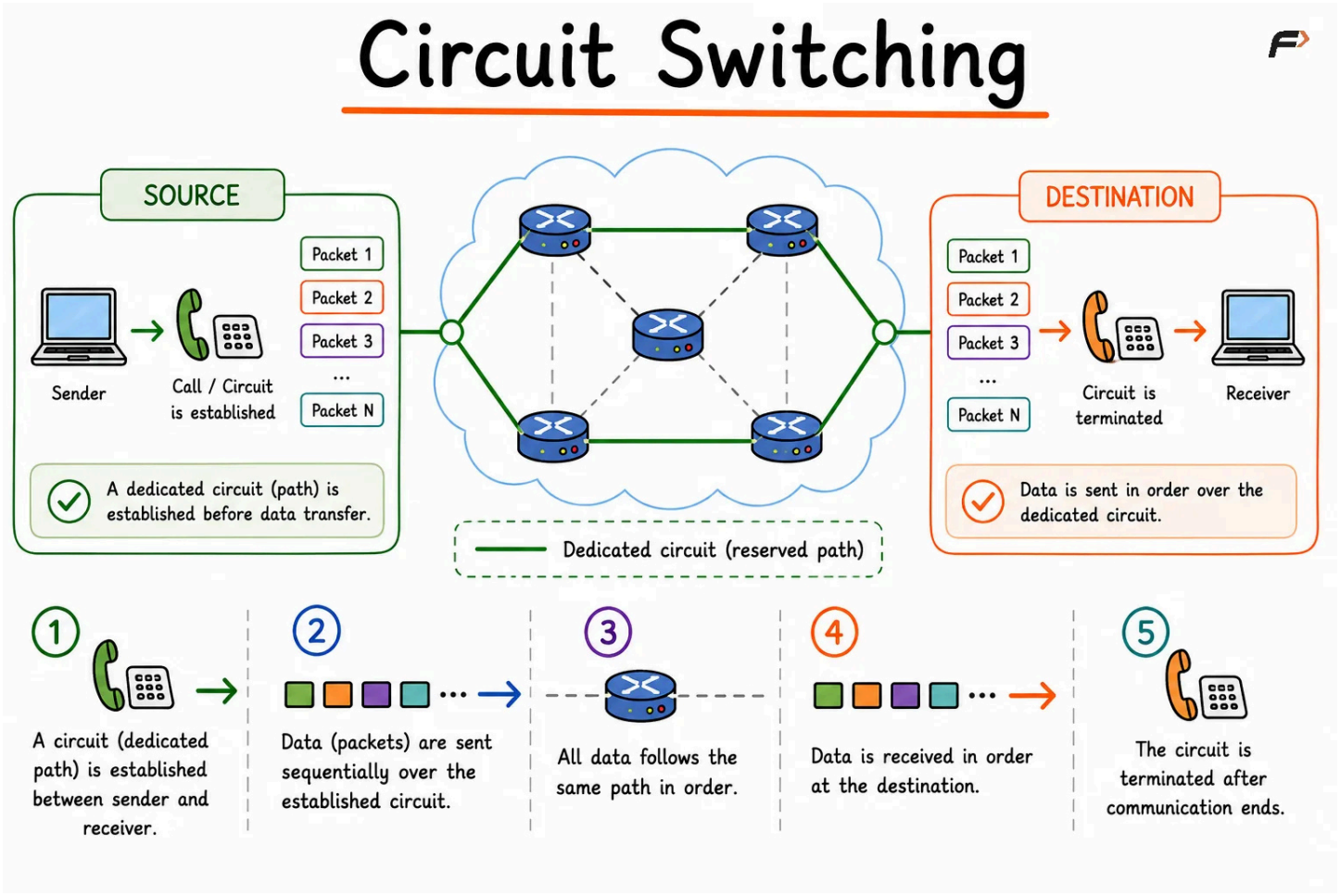
Packets may take different routes.

Network resources are shared among multiple users.

High resource utilization and efficiency.

Circuit Switching

Circuit switching establishes a dedicated communication path between the sender and receiver before data transmission begins. The path remains reserved throughout the communication session.



Key characteristics include:

A dedicated path is created before communication starts.

Resources remain reserved for the duration of the connection.

Data follows the same route throughout the session.

Communication is continuous and predictable.

Why the Internet Uses Packet Switching

Internet traffic is typically bursty in nature. Users do not continuously transmit data. Instead, communication happens in short bursts followed by periods of inactivity.

A typical web interaction looks like this:

User clicks a link.

A small request is sent.

The user waits while the server processes the request.

A burst of response data is received.

The user pauses before the next action.

If circuit switching were used, network resources would remain reserved even during idle periods, resulting in significant wastage. Packet switching avoids this problem by allowing multiple users to share the same network infrastructure efficiently.

Comparison

Aspect	Packet Switching	Circuit Switching
Data Transmission	Data is broken into packets.	Data flows as a continuous stream.
Path	No dedicated path.	Dedicated path established.
Resource Usage	Shared among users.	Reserved for one connection.
Efficiency	High.	Lower during idle periods.
Example	Internet	Traditional telephone networks

Why Layered Architecture?

This module is about the **layered design of computer networks**. Without understanding this, networking often feels like a collection of disconnected protocols rather than a structured system.

Let’s assume your laptop sends a message to a website server. On the surface it feels simple, you send data and receive a response. But internally, multiple responsibilities must be handled in a coordinated way:

The application has to create meaningful data such as messages, emails, files, or HTTP requests.

The data may need formatting, compression, or encryption.

Reliability may be required (ordering, retransmission, flow control).

Data has to be routed across many networks.

On each local link, devices must transmit electrical, optical, or radio signals.

Hardware from different vendors and networks run by different organizations still need to interoperate.

Each of these responsibilities is fundamentally different in nature. Some deal with application logic, some with reliability, some with routing, and some with physical transmission.

For example, data may travel from your laptop to a home router, then to an ISP router, through multiple networks, and eventually reach the destination server. Every stage has different requirements and challenges.

If one giant protocol tried to handle all of this, it would become extremely difficult to design, maintain, replace, debug, and standardize.

When something goes wrong, identifying the source of the problem would also become extremely difficult. Troubleshooting would require understanding every part of the communication process at once.

What is a Protocol?

A **protocol** is a set of communication rules between computers. It defines how data is formatted, transmitted, and interpreted.

Examples include:

HTTP for requesting and transferring web content.

TCP for reliable communication.

IP for addressing and routing packets across networks.

Different protocols solve different networking problems. Together, they enable end-to-end communication across the Internet.

Need for Layered Architecture

A **layered architecture** divides networking into separate responsibilities, where each layer focuses on a specific task.

This is very similar to software modularity, where complex systems are broken into smaller, manageable components.

Instead of one large system handling everything, each layer solves a specific problem and interacts with the layers above and below it.

How Layers Work

Think of networking as a stack of responsibilities:

The **browser** cares about HTTP requests and responses.

TCP cares about delivery, ordering, and retransmission.

IP cares about routing packets across networks.

The **physical layer** cares about transmitting bits over wires, fiber, or wireless media.

Each layer performs its own responsibility and then passes the result to another layer.

A layer uses the services provided by the layer below it and provides services to the layer above it. In this way, all layers cooperate to achieve end-to-end communication.

Each layer does not need to know how the other layers are implemented internally. It only needs to follow a defined interface.

For example, the browser does not need to know how TCP ensures reliability, and TCP does not need to know how signals are transmitted over fiber or Wi-Fi.

Why Layers are Powerful

Modularity: Each layer has a well-defined responsibility, making systems easier to design, maintain, and extend.

Interoperability: Devices and software from different vendors can communicate as long as they follow common protocol standards.

Easier Upgrades: Changes in one layer can often be made without affecting the entire networking stack.

Easier Debugging: Problems can be isolated to a specific layer, making troubleshooting much simpler.

Standardization: Well-defined layers and protocols allow consistent implementations across the world.

Summary

Networking is not one single system. Instead, it is a **stack of cooperating responsibilities**.

Each layer solves a specific problem, uses the services of the layer below it, and provides services to the layer above it.

This layered approach makes networks easier to design, standardize, maintain, upgrade, and debug. Without layering, building and operating modern computer networks, especially the Internet, would be extremely difficult.